Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	06-april-2026
Team ID	SWTID-2026-2633
Project Name	Automating import data and relationship mapping using import sets and dot walking
Maximum Marks	4 Marks

Technology Stack

Key Layers:

- Data Sources: ERP, HRMS, CSV, APIs
- Integration Layer: ETL / Middleware
- Staging Layer: Import Sets
- Transformation Layer: Transform Maps + Scripts
- Data Layer: CMDB / Application Tables
- Consumption Layer: Reports, APIs, UI

2. Core Platform Stack

Platform Layer

- ServiceNow
  - Import Sets
  - Transform Maps
  - CMDB (Configuration Management Database)
  - Flow Designer
  - Business Rules
  - Script Includes (server-side logic)

3. Integration Layer

Middleware / iPaaS

- MuleSoft
- Dell Boomi
- Azure Logic Apps

API Communication

- REST APIs (JSON/XML)
- SOAP (legacy systems)

Data Transfer

- SFTP / FTP
- Webhooks
- Scheduled pulls

4. Data Storage & Staging

Staging Layer

- Import Set Tables (temporary staging tables)
- Raw, unprocessed data storage

Target Data Layer

- CMDB Tables (e.g., `cmdb_ci`)
- Custom Application Tables

5. Transformation & Processing

Transform Engine Components

- Transform Maps
- Field Maps
- Coalesce (for deduplication)

- Scripted transforms (JavaScript)

Logic Handling

- Data normalization
- Relationship creation
- Reference resolution (foreign keys)

6. Relationship Mapping & Dot Walking

Relationship Handling

- Reference fields (`sys_id`)
- CMDB relationship tables (`cmdb_rel_ci`)
- Lookup rules

Dot Walking Usage

- Access related fields (e.g., `user.manager.department`)
- Used in:
    - Scripts
    - UI policies
    - Reports
    - Transform logic

7. Automation Layer

Workflow & Automation

- Flow Designer (low-code automation)
- Scheduled Jobs (cron-based imports)
- Event-driven triggers

8. Scripting & Logic Layer

Languages

- JavaScript (server-side & client-side)

Components

- Business Rules
- Script Includes
- Transform Scripts
- UI Scripts

9. Monitoring & Logging

Logging

- Import Set Run History
- Transform History
- Error Tables

Observability Tools

- Platform logs
- External monitoring:
    - Splunk
    - ELK Stack

10. Security Layer

- Role-Based Access Control (RBAC)
- Data encryption (in transit + at rest)
- API authentication:
    - OAuth 2.0
    - Basic Auth

- Field-level security

11. Cloud & Infrastructure

Hosting Platforms

- Amazon Web Services
- Microsoft Azure

Storage & Compute

- Managed databases
- Serverless functions (optional preprocessing)

12. CI/CD & DevOps

Version Control

- Git

CI/CD Pipelines

- Jenkins
- GitHub Actions

ServiceNow Deployment

- Update Sets
- Application Repository

13. Example End-to-End Stack

Scenario: HR Data Import into CMDB

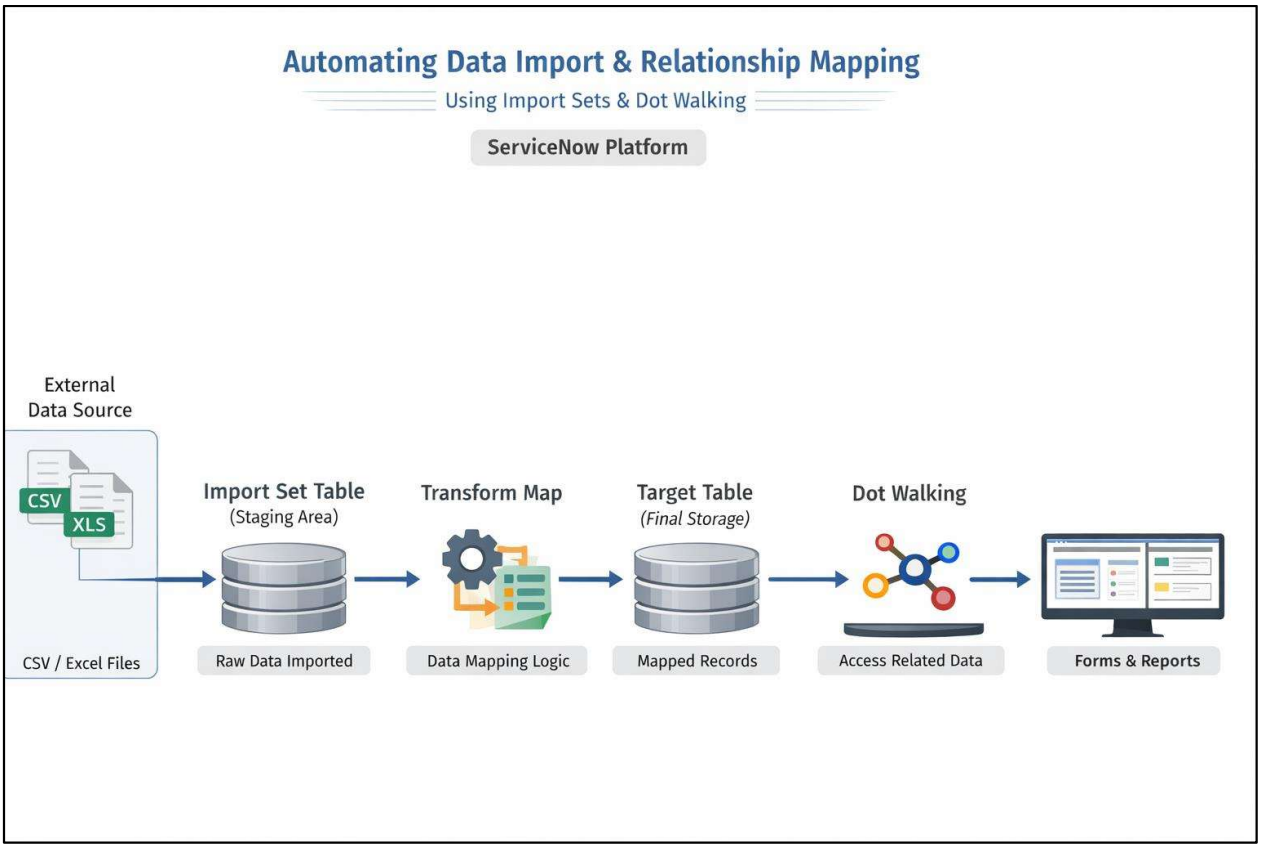
- Source: HRMS (via REST API)

- Integration: MuleSoft
- Platform: ServiceNow
- Staging: Import Set Table
- Transform: Transform Map + Script
- Mapping:
  - Employee → User table
  - Manager → Reference field
- Relationship:
  - Employee → Department
- Automation: Scheduled job (daily sync)
- Monitoring: Splunk

14. Architecture Patterns

Common Patterns

- ETL (Extract, Transform, Load)
- ELT (for large-scale data)
- Event-driven architecture
- Batch vs Real-time processing


Automating Data Import & Relationship Mapping
Using Import Sets & Dot Walking
ServiceNow Platform
External Data Source
CSV
XLS
CSV / Excel Files
Import Set Table
(Staging Area)
Raw Data Imported
Transform Map
Data Mapping Logic
Target Table
(Final Storage)
Mapped Records
Dot Walking
Access Related Data
Forms & Reports